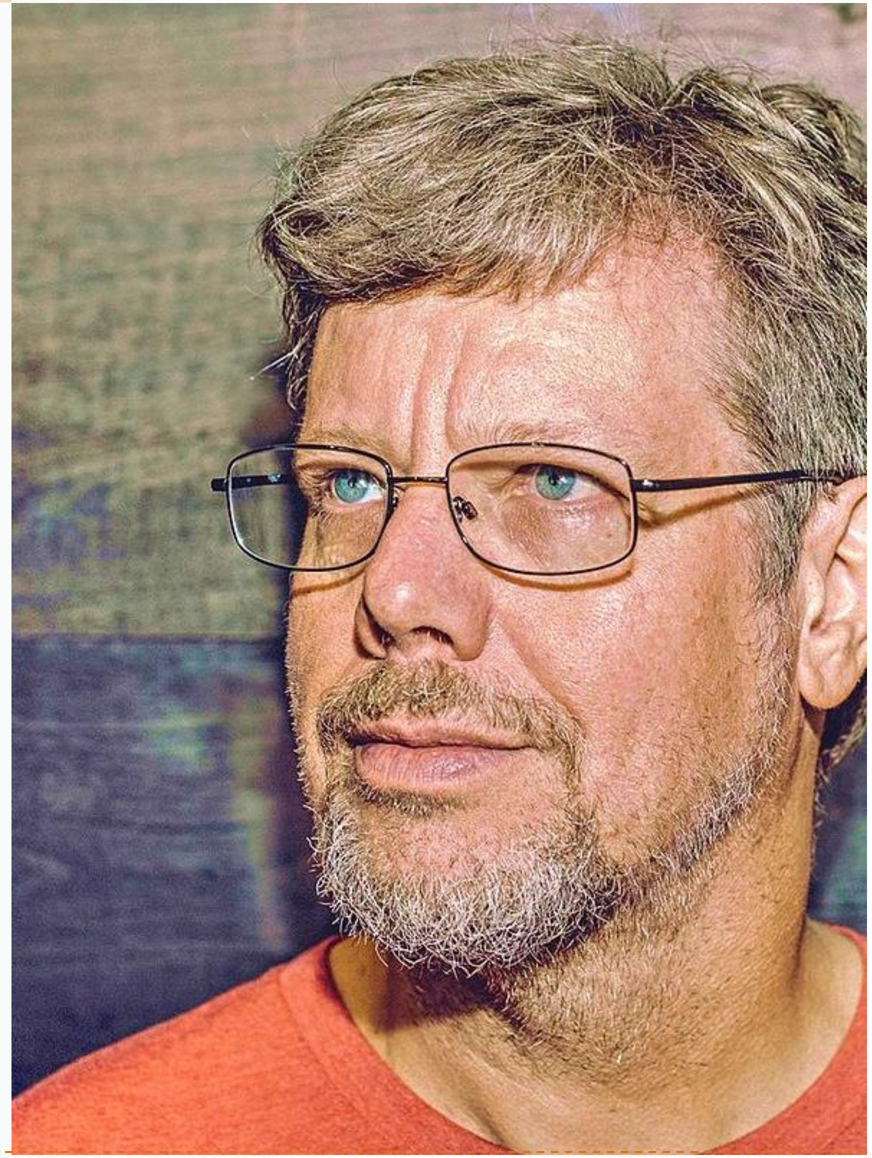
PYTHON INTRODUCTION

Topics:

- History
- Features
- Operators in Python



History

Python was conceived in the late 1980 by Guido van Rossum at Centrum Wiskunde & Informatica (CWI) in the Netherlands as a successor to ABC programming language, which was inspired by SETL.

Python 2.0 was released on 16 October 2000 with many major new features, including a cycle-detecting garbage collector and support for Unicode.

late 1980

Dec. 1989

16 Oct. 2000

3 Dec. 2008

Its implementation began in December 1989.

Python 3.0 was released on 3 December 2008

Naming

Python's name is derived from the British comedy group Monty Python, whom Python creator Guido van Rossum enjoyed while developing the language. Monty Python references appear frequently in Python code and culture.

Features

Python is a dynamic, high level, free open source and interpreted programming language. It supports object-oriented programming as well as procedural oriented programming.
In Python, we don’t need to declare the type of variable because it is a dynamically typed language.
For example, x = 10
Here, x can be anything such as String, int, etc.

Features

Easy to code:
Python is a high-level programming language. Python is very easy to learn the language as compared to other languages like C, C#, Java script, Java, etc. It is very easy to code in python language and anybody can learn python basics in a few hours or days. It is also a developer-friendly language.

Features

Free and Open Source:
Python language is freely available at the official website and you can download it from the given download link below www.python.org

Since it is open-source, this means that source code is also available to the public. So you can download it as, use it as well as share it.

Features

Object-Oriented Language:
One of the key features of python is Object-Oriented programming. Python supports object-oriented language and concepts of classes, objects encapsulation, etc.

Features

GUI Programming Support:
Graphical User interfaces can be made using a module such as PyQt5, PyQt4, wxPython, or Tk in python.
PyQt5 is the most popular option for creating graphical apps with Python.

High-Level Language:
Python is a high-level language. When we write programs in python, we do not need to remember the system architecture, nor do we need to manage the memory.

Features

Python is Integrated language:
Python is also an Integrated language because we can easily integrate python with other languages like c, c++, etc.

Python is Portable language:
Python language is also a portable language. For example, if we have python code for windows and if we want to run this code on other platforms such as Linux, Unix, and Mac then we do not need to change it, we can run this code on any platform.

Features

Interpreted Language:

Python is an Interpreted Language because Python code is executed line by line at a time. like other languages C, C++, Java, etc. there is no need to compile python code this makes it easier to debug our code. The source code of python is converted into an immediate form called **bytecode**.

Features

Large Standard Library

Python has a large standard library which provides a rich set of module and functions, so you do not have to write your own code for everything. There are many libraries present in python for such as regular expressions, unit-testing, web browsers, etc.

Operators

Arithmetic Operators

Operator	Name	Example
+	Addition	x + y
-	Subtraction	x – y
*	Multiplication	x * y
/	Division	x / y
%	Modulus	x % y
**	Exponentiation	x ** y
//	Floor Division	x // y

Operators

Comparison Operators

Operator	Name	Example
==	Equal	x == y
!=	Not Equal	x != y
>	Greater than	x > y
<	Less Than	x < y
>=	Greater than or Equal to	x >= y
<=	Less than or Equal to	x <= y

Operators

Logical Operators

Operator	Name	Example
and	Returns True if both statements are true	x < 5 and x < 10
or	Returns True if one of the statements is true	x < 5 or x < 4
not	Reverse the result, returns False if the result is true	not x < 5